

West University of Timisoara

Master's Program

Specialization: Big Data – Data Science, Analytics and Technologies

Course: Data Mining

Email Spam Detection

using PCA, Feature Selection, Naive Bayes and SVM

Student:

Ioan Bradesa

Abstract

This project addresses the problem of automatic email spam detection using the UCI Spambase dataset. The objective is to classify emails as spam or non-spam based on numerical features extracted from email content. The dataset contains word frequency attributes, character frequency attributes and capital letter sequence statistics. The classification task is relevant because spam emails can waste user time, increase storage and processing costs, and may also represent security risks through phishing, fraud or malicious links.

The implemented processing workflow includes exploratory data analysis, statistical description of the dataset, feature scaling, feature selection, dimensionality reduction with Principal Component Analysis, and supervised classification. Two classification algorithms were evaluated: Gaussian Naive Bayes and Support Vector Machine with RBF kernel. Feature selection was performed using SelectKBest with ANOVA F-score, while PCA was used to preserve approximately 95% of the variance in the data.

The best result was obtained by the baseline SVM model, with an accuracy of 92.73%, precision of 92.77%, recall of 88.43%, F1-score of 90.55% and ROC-AUC of 96.67%. SVM with PCA achieved very similar performance, showing that dimensionality reduction preserved most of the useful information. Naive Bayes also performed reasonably well, but it was less balanced than SVM.

1 Introduction

Email spam detection is a classical binary classification problem in machine learning. The task consists of deciding whether a given email is spam or non-spam based on measurable characteristics extracted from the message. Spam emails are commonly associated with advertising, scams, phishing attempts, suspicious links and other unwanted content. Automatic spam filtering is therefore important for both productivity and cybersecurity.

The dataset selected for this project is the UCI Spambase dataset. It is suitable for a data mining project because it is a real dataset, publicly available, already structured in numerical format and designed specifically for classification. The dataset contains 4601 email samples and 57 input features. The target variable indicates whether an email is spam or not spam. According to the UCI Machine Learning Repository, the spam class includes emails such as advertisements, chain letters and other unsolicited messages, while the non-spam class includes personal and work-related emails.

Previous studies on spam detection have commonly used probabilistic models, feature-based representations and supervised classification algorithms. Naive Bayes has been widely used for spam filtering because of its simplicity, speed and probabilistic interpretation. Androutsopoulos et al. evaluated Naive Bayes for anti-spam filtering and discussed its advantages and limitations, especially the need to consider false positives in practical systems [2]. Support Vector Machines have also been used frequently in text classification and spam detection because they can separate classes effectively in high-dimensional spaces. Recent spam classification research still compares traditional methods such as Naive Bayes, SVM and other classifiers, often using feature extraction or dimensionality reduction techniques [3].

The solution implemented in this project compares two classifiers: Gaussian Naive Bayes and Support Vector Machine. The workflow also includes feature selection and Principal Component Analysis in order to study whether reducing the number of features improves or preserves classification performance. The report is structured as follows: Section 2 describes the dataset and its statistical characteristics, Section 3 presents the processing flow, Section 4 discusses the results, and Section 5 gives the conclusions.

2 Description of the Dataset

The dataset used in this project is the Spambase dataset from the UCI Machine Learning Repository. Each record represents one email. The classification objective is to determine whether the email is spam or non-spam.

The dataset contains 4601 records and 58 columns in total. Out of these, 57 columns

are input features and one column is the target variable. The target variable is binary:

$$0 = \text{non-spam}, \quad 1 = \text{spam}$$

The input attributes are numerical and can be grouped into three main categories:

- word frequency attributes, such as `word_freq_free`, `word_freq_money`, `word_freq_remove`;
- character frequency attributes, such as `char_freq_!` and `char_freq_$`;
- capital letter sequence attributes, such as `capital_run_length_average`, `capital_run_length_longest` and `capital_run_length_total`.

The class distribution is presented in Table 1.

Table 1: Class distribution of the Spambase dataset

Class	Number of emails	Percentage
Non-spam	2788	60.60%
Spam	1813	39.40%

The dataset is moderately imbalanced, because the non-spam class is larger than the spam class. However, the imbalance is not extreme. Because of this, stratified train-test splitting was used in the implementation to preserve the same class proportions in both the training and test sets.

No missing values were found in the dataset. All input attributes are numerical, so no categorical encoding was required. Descriptive statistics showed that many features are strongly skewed and contain many values close to zero. This is expected because most emails do not contain every selected word or special character. Some attributes, especially the capital letter sequence features, contain larger extreme values. Because of this, feature scaling was necessary before applying SVM and PCA.

Selected descriptive statistics for relevant features are presented in Table 2.

Table 2: Descriptive statistics for selected features

Feature	Mean	Std. Dev.	Min	Max
<code>word_freq_free</code>	0.2488	0.8258	0.0000	20.0000
<code>word_freq_money</code>	0.0943	0.4426	0.0000	12.5000
<code>word_freq_remove</code>	0.1142	0.3914	0.0000	7.2700
<code>word_freq_you</code>	1.6621	1.7755	0.0000	18.7500
<code>char_freq_exclamation</code>	0.2691	0.8157	0.0000	32.4780
<code>char_freq_dollar</code>	0.0758	0.2459	0.0000	6.0030
<code>capital_run_length_average</code>	5.1915	31.7294	1.0000	1102.5000
<code>capital_run_length_longest</code>	52.1728	194.8913	1.0000	9989.0000
<code>capital_run_length_total</code>	283.2893	606.3479	1.0000	15841.0000

The exploratory analysis also included histograms and boxplots. The histograms showed that most features have right-skewed distributions. The boxplots indicated that some words and characters, such as “free”, “money”, “remove”, “!” and “\$”, tend to appear more frequently in spam emails than in non-spam emails. This suggests that these features are useful for classification.

3 Description of the Processing Flow

The implemented workflow was developed in Python using Google Colab. The main libraries used were Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn.

The first step was loading the dataset directly from the UCI Machine Learning Repository. Since the original data file does not contain column names, the official Spambase attribute names were added manually. After loading the dataset, basic inspection was performed, including the number of records, number of attributes, class distribution and missing value analysis.

The second step was exploratory data analysis. Descriptive statistics were computed for all attributes, and additional statistics were examined for selected features considered relevant for spam detection. Visualizations were also created, including a class distribution plot, histograms for selected features and boxplots comparing spam and non-spam emails.

The third step was splitting the dataset into input variables and target variable. The 57 input features were stored in matrix X , while the target variable was stored in vector y . The dataset was split into 80% training data and 20% testing data. Stratified sampling was used in order to preserve the original class proportions in both subsets. The training set contained 3680 samples, while the test set contained 921 samples.

Feature scaling was then applied using StandardScaler. Scaling was necessary because SVM and PCA are sensitive to the scale of the input variables. StandardScaler transforms the attributes so that they have approximately zero mean and unit variance based on the training data.

The first models trained were baseline models. Gaussian Naive Bayes was selected because it is simple, fast and commonly used in spam detection tasks. Support Vector Machine with RBF kernel was selected because SVM is effective for binary classification and can model non-linear class boundaries.

After the baseline models, feature selection was applied. SelectKBest with ANOVA F-score was used to select the 20 most informative features. This method evaluates each feature individually and assigns a score based on how well it separates the classes. The selected features were then used to train Naive Bayes and SVM again.

Finally, Principal Component Analysis was applied as a dimensionality reduction method. PCA was performed after standardization and was configured to preserve 95% of the variance. This reduced the dataset from 57 original features to 48 principal com-

ponents while preserving approximately 95.49% of the variance.

All models were evaluated using the same test set. The performance metrics used were accuracy, precision, recall, F1-score and ROC-AUC. Confusion matrices and classification reports were also generated.

4 Presentation of Results

The final comparison of the trained models is presented in Table 3. The models are ordered by F1-score.

Table 3: Final model comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
SVM baseline	0.9273	0.9277	0.8843	0.9055	0.9667
SVM + PCA	0.9262	0.9275	0.8815	0.9040	0.9679
SVM + Feature Selection	0.9207	0.9119	0.8843	0.8979	0.9628
Naive Bayes baseline	0.8328	0.7146	0.9587	0.8188	0.9376
Naive Bayes + PCA	0.8339	0.7323	0.9118	0.8123	0.9256
Naive Bayes + Feature Selection	0.8502	0.8814	0.7163	0.7903	0.9244

The best overall model was the baseline SVM. It achieved an accuracy of 92.73%, precision of 92.77%, recall of 88.43%, F1-score of 90.55% and ROC-AUC of 96.67%. This shows that SVM was able to obtain a strong balance between correctly identifying spam emails and avoiding false positives.

SVM with PCA achieved very similar performance. Its accuracy was 92.62% and its F1-score was 90.40%. Although it did not improve over the baseline SVM in terms of F1-score, it reduced the dimensionality from 57 original features to 48 components while preserving approximately 95.49% of the variance. This shows that PCA preserved most of the useful information from the original feature space.

SVM with feature selection also performed well, with an accuracy of 92.07% and F1-score of 89.79%. This model used only 20 selected features instead of all 57 original features. Therefore, even though the performance was slightly lower than the baseline SVM, the result shows that a smaller subset of informative attributes can still provide strong classification performance.

The Naive Bayes baseline achieved an accuracy of 83.28% and an F1-score of 81.88%. It had very high recall, equal to 95.87%, meaning that it detected most spam emails. However, its precision was lower, equal to 71.46%, which means that it produced more false positives than SVM. In a spam detection system, false positives are important because incorrectly marking legitimate emails as spam can be undesirable.

Naive Bayes with feature selection improved accuracy compared to the Naive Bayes baseline, from 83.28% to 85.02%. However, recall decreased from 95.87% to 71.63%, which

means that the model missed more spam emails. Naive Bayes with PCA achieved similar accuracy to the baseline but did not outperform SVM.

Compared with existing literature, the obtained results are consistent with the general observation that SVM is often a strong classifier for text-related classification tasks. Previous spam filtering studies have shown that Naive Bayes is simple and efficient, but it may require careful handling of false positives and feature representation [2]. More recent spam classification work has also shown that SVM-based models can achieve strong results when combined with suitable text representations [3]. In this project, SVM also achieved the best overall performance on the Spambase dataset.

5 Conclusions

This project implemented and evaluated several machine learning workflows for email spam detection using the UCI Spambase dataset. The processing flow included dataset inspection, statistical analysis, visualization, feature scaling, feature selection, PCA dimensionality reduction and classification using Naive Bayes and SVM.

The best result was obtained by the baseline SVM model. It achieved the highest F1-score and strong values for accuracy, precision, recall and ROC-AUC. This indicates that SVM was the most suitable classifier among the methods tested in this experiment.

Feature selection reduced the number of attributes from 57 to 20 and preserved good performance for SVM, but it did not improve over the baseline SVM. PCA reduced the feature space from 57 original features to 48 principal components while preserving approximately 95.49% of the variance. The SVM model trained on PCA components performed almost the same as the baseline SVM.

Naive Bayes was faster and simpler, but its performance was less balanced. The baseline Naive Bayes model had high recall but lower precision, meaning that it detected many spam emails but also produced more false positives. Overall, the results show that SVM provides the most reliable performance for this dataset.

Possible future improvements include testing additional classifiers such as Random Forest, Logistic Regression or Gradient Boosting, using cross-validation instead of a single train-test split, and exploring other feature selection methods.

References

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